the group determinant is equal to the index of the commutator subgroup and admitted, "I distinctly feel that I will not achieve anything here" (quoted in [**24**, pp. 223, 224]).

Dedekind's challenge was perfectly timed for the spring break, and Frobenius, having some free time on his hands, immediately took the bait. Employing military metaphors to describe his own work, he waged a war against the group determinant. He invented generalized group characters and assaulted Dedekind's conjecture with their help. Frobenius's initial results seemed unsatisfactory, and he reported them to Dedekind with a disclaimer that "my conclusions are so complicated that I myself do not rightly know where the main point of the proof is, and in fact I am still slightly mistrustful of it" (quoted in [**24**, pp. 225, 230]).

A few days later a jubilant Frobenius wrote to Dedekind that he finally saw the way to a solution. Citing his former colleague Friedrich Schottky, Frobenius now looked at his own earlier frustration as the harbinger of a forthcoming breakthrough: "If in an investigation, after difficult mental exertion, the feeling arises that nothing will be achieved on the matter in question, then one can rejoice for he is standing before the solution" (quoted in [**24**, p. 230]). Within ten days, Frobenius proved all the main theorems of the theory of group determinants, except for one most important result.

The missing theorem stated, in modern language, that an irreducible representation occurs in the regular representation as often as its degree and was called by Frobenius "the Fundamental Theorem of the theory of group determinants". "It would be wonderful" if it were true, he wrote, "for then my theory would supply everything needful" for the determination of prime factors (quoted in [**24**, p. 235]). The proof took five months, during which Frobenius repeatedly used a new, effective technique, which he kept secret and shared privately only with Dedekind. The new method drew on the well-known bargaining strategy: "At the market, the desired horse is ignored as much as possible and at last is allowed to be formally recognized" (quoted in [**24**, p. 236]). In Frobenius's interpretation, in order to solve a mathematical problem, one had to preoccupy oneself with activities totally unrelated to mathematics. He applied this "Principle of the